
Toward Auditable Vision-Language Control

A Fully Tensor-Decomposable VLA

Specialist capability and exact weight-level diagnostics

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ -VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the convolutional 19.2M-parameter instantiation reaches $93.7\% \pm 0.8\%$ under the published three-seed protocol. Averaging its three fixed checkpoints reaches 96.2% at three times the inference cost. On the three released convolutional checkpoint files associated with those historical aggregates, independent reconstruction covers all 4,608 joint-attention head-input cases with worst relative error 6.36×10^{-16} . In a separate three-checkpoint ViT instantiation, we derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module with worst relative error 2.56×10^{-7} . All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. The evidence establishes strong specialist capability together with exact, checkpoint-native access to learned attention structure. Such access is relevant when deployment failures may involve visual shortcuts or weak language use. The present results do not establish broad robotic generalization, robust instruction grounding, selective behavioral editing, or an exact end-to-end decomposition of the full policy.

1 Introduction

Deployable vision-language robot policies need more than aggregate success. When a policy fails, developers need to distinguish possible visual shortcuts, weak instruction use, state-estimation errors, and unstable action generation. Conventional checkpoints expose little of this structure directly. Post-hoc probes can identify correlated activations, but their conclusions depend on a chosen dataset and need not reveal how the deployed weights construct an action.

Tensor decomposition offers a complementary, checkpoint-native analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic

form. This structure does not certify semantic grounding by itself. It provides a direct weight-level surface for subsequent grounding and failure analyses.

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. Exact structural access is useful only if the restricted policy still controls reliably.

We therefore study a prerequisite for deployment-oriented diagnostics:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while exposing exact layerwise structure from its weights?

We test capability and structural access in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit. These experiments test whether a useful diagnostic surface can coexist with control, not whether that surface already certifies grounding or supports a validated edit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate fully rational 19.2M convolutional and 20.1M ViT policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. Our narrower result is that an algebraic VLA can make exact layerwise weight structure directly accessible without giving up strong specialist control.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

76 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

77 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value,
78 and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike
79 softmax attention, every multiplicative interaction can be expanded into coefficients determined by
80 the trained weights.

81 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
82 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
83 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

84 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
85 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
86 operations update this pair. The model therefore keeps input-specific rescaling while remaining
87 exactly representable as a rational tensor network.

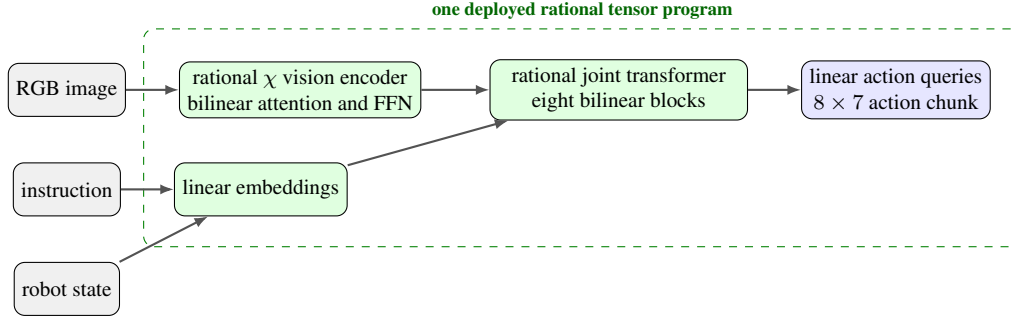


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

88 2.2 Architecture and training

89 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
90 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
91 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
92 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
93 continuous action dimensions. The complete model has 20,137,352 learned parameters.

94 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
95 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
96 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
97 that verify every requested state was loaded.

98 2.3 Mechanical certificate on the 20.1M ViT checkpoint

99 Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT
100 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

Mechanical decomposability audit of the trained 189/200 checkpoint

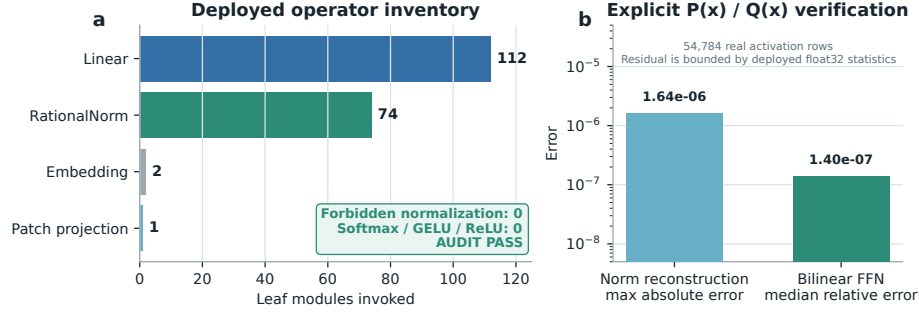


Figure 2: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

101 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 102 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 103 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does
 104 not materialize one compact polynomial for the full eight-layer transformer.

105 3 Protocol-aligned closed-loop capability

106 3.1 Evaluation protocol

107 We evaluate all ten LIBERO-Object tasks with:

- 108 • 50 canonical trials per task
- 109 • a 280-step cap
- 110 • three independently trained seeds per architecture
- 111 • 500 rollouts per seed and 1,500 rollouts per architecture

112 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 113 rerun in our codebase, so differences may still include implementation and training choices.

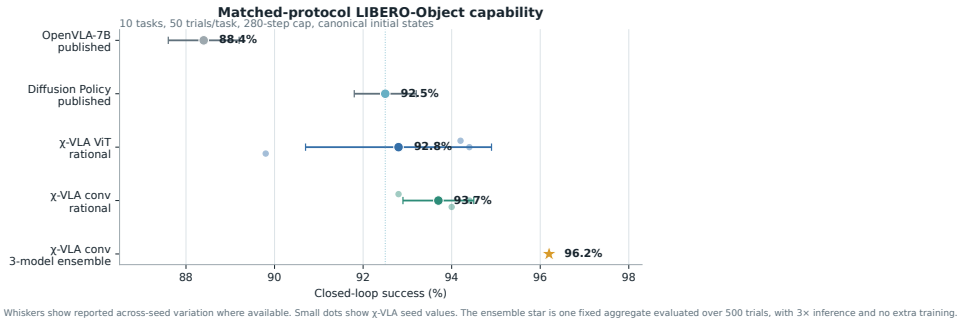


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

114 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	19.2M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 19.2\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

115 The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow
116 spread is descriptive bounded evidence across independently trained checkpoints.

117 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and
118 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points
119 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
120 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
121 19.2M-parameter policy.

122 3.3 What this result does and does not establish

123 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
124 existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization
125 or state of the art.

126 4 Exact weight-derived structure through attention

127 4.1 Exact layerwise construction

128 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
129 a high-order coefficient table. Each entry says how strongly one combination of query, key, and
130 value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe
131 are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and
132 an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to
133 1.24×10^{-14} .

134 The deployed attention also normalizes each query and key with Equation 3. Each normalization
135 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
136 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
137 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
138 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
139 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
140 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
141 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

142 We separately certify the three released convolutional checkpoint files associated with the historical
143 capability aggregates in Section 3. Their vision stacks contain no attention, so we independently
144 rebuild all eight joint modules and all twelve heads per module from the saved weights. Sixteen
145 deterministically selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input
146 cases. Every output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6}
147 gate. This same-checkpoint certificate covers the joint-attention equation at each normalized layer
148 input. It does not rebuild the convolutional stem, feed-forward blocks, residual composition, final
149 normalization, or action head as one end-to-end contraction.

150 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
151 values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays
152 the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most
153 coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force
154 materialization to 2.13×10^{-14} .

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer’s computation.

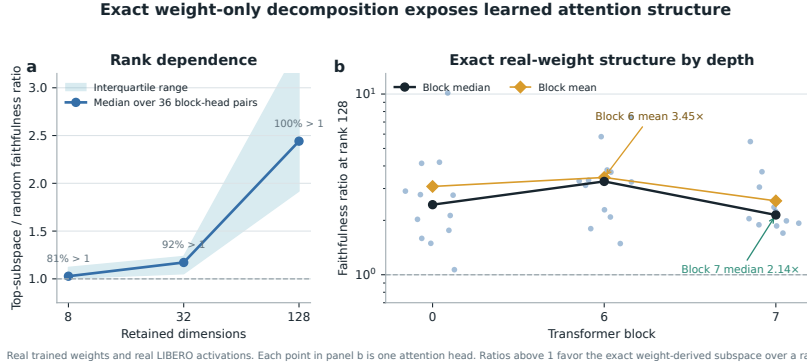


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight auditing in a strong continuous vision-language policy.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. This motivates model-internal diagnostics, but our exact weight

analysis does not itself certify grounding. We therefore treat LIBERO-Object as a bounded specialist capability setting rather than evidence of broad deployment readiness.

6 Limitations and conclusion

χ -VLA demonstrates that strong specialist closed-loop control can coexist with a mechanically auditable rational graph and exact layerwise attention analysis. This combination is relevant to deployment diagnostics because it exposes trained coefficient structure without fitting a separate probe.

The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions. The experiments do not show that the policy follows language when visual cues conflict, generalizes across task distributions, or transfers to physical robots. Exact reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

We therefore present χ -VLA as diagnostic infrastructure, not as a deployment-ready or robustly grounded robot policy. The released convolutional files support the joint-attention equation certificate and are associated with the historical capability aggregates. Separate ViT checkpoints support the reduced-Gram subspace screen. Broader suites, counterfactual instruction tests, and physical-robot evaluation remain necessary for deployment claims. We do not claim a direct coefficient-edit interface.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

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A Supporting results and decisions

A.1 Per-task ensemble behavior

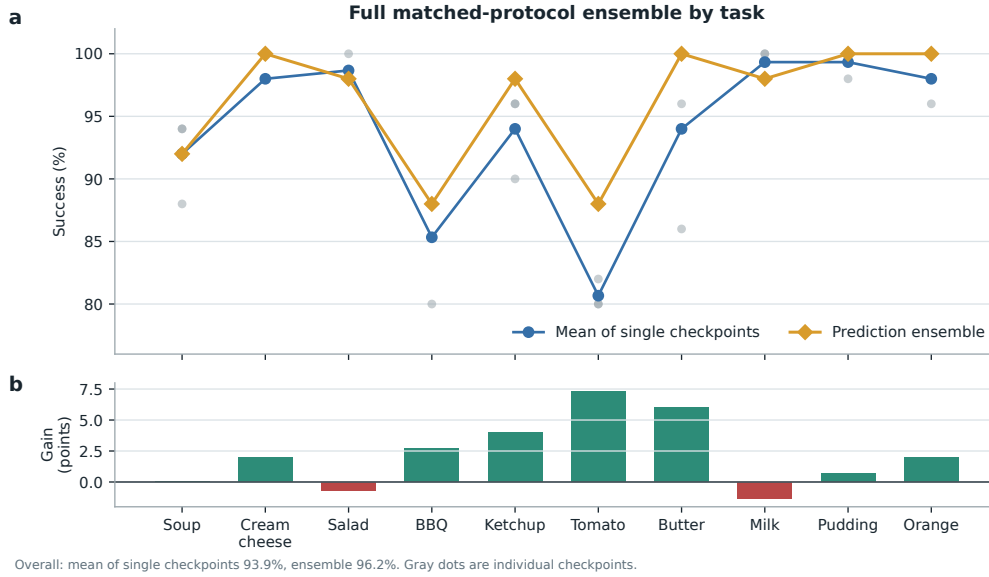


Figure 5: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

A.2 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

247 A.3 Numerical scope of rational conversion

248 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 249 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 250 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 251 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 252 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 253 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and
 254 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that
 255 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined
 256 as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3}
 257 gate.

258 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 259 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 260 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 261 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 262 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 263 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 264 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

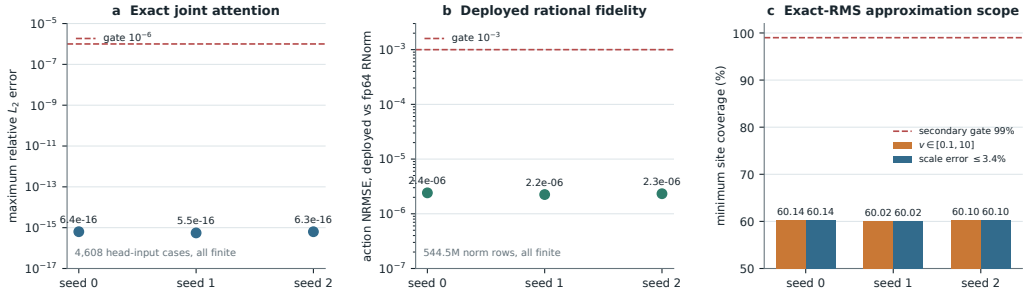


Figure 6: **Frozen structural certificates on the released convolutional files associated with the historical capability aggregates.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.